

Def. Let X be a set, $(Y, \mathcal{U})$ be a Topological Space.

For a map $f: X \rightarrow Y$, define:

$$\mathcal{U}^f \stackrel{\text{def}}{=} \{f^{-1}[U] \subseteq X \mid U \in \mathcal{U}\}$$

It is called as generated Topology on X by f .

Prop. Lower Limit Topology on $\mathbb{R}$ is Not Second-Countable.

Proof. Suppose that $(\mathbb{R}, \tau_L)$ has a Countable basis β .

Then, for any $x \in \mathbb{R}$, there exists a $B_x \in \beta$ s.t. $x \in B_x \subseteq [x, x+1)$.

Clearly, if $x < y$, $B_x \neq B_y$. Now, the map $\mathbb{R} \rightarrow \beta; x \mapsto B_x$ is an injective,

Thus $\#(\beta) > \aleph_0$, which is contradiction. $\square$

Prop. Co-Finite Topology on $\mathbb{R}$ is Not Second-Countable.

Proof. Let β be a Countable basis for $\mathcal{C}_f$ not containing $\emptyset$.

(If β has $\emptyset$, $\beta \setminus \{\emptyset\}$ is still a basis which generated same Topology.)

Now, $\mathbb{R} \neq \bigcup_{B \in \beta} B$ because $\mathbb{R} \setminus B$ is finite and β is countable.

Let $x \in \mathbb{R} \setminus \left(\bigcup_{B \in \beta} B\right)$. Then, clearly $\mathbb{R} \setminus \{x\}$ is open of $\mathcal{C}_f$.

But, $x \in \mathbb{R} \setminus \bigcup_{B \in \beta} B$ implies that $\forall B \in \beta, x \notin B$.

By definition, for any $p \in \mathbb{R} \setminus \{x\}$, there exists $B \in \beta$ s.t.

$$p \in B \subseteq \mathbb{R} \setminus \{x\}$$

, which means $\{x\} \subseteq B$. Contradiction. $\square$

Note: Similarly, Co-countable space on $\mathbb{R}$ is Not Second-Countable.

Thm. Let β be a basis on a set X . Then,

$$\mathcal{C}_\beta = \bigcap_{\substack{\beta \subseteq \mathcal{C} \\ \mathcal{C} \text{ is Topology}}} \mathcal{C}$$

Proof. Let $U \in \mathcal{C}_\beta$ be given. Then, for some $\beta' \subseteq \beta$, $U = \bigcup_{B \in \beta'} B$.

Clearly, $\beta \subseteq \bigcap_{\substack{\beta \subseteq \mathcal{C} \\ \mathcal{C} \text{ is Topology}}} \mathcal{C}$. Thus, $U = \bigcup_{B \in \beta'} B \in \bigcap_{\substack{\beta \subseteq \mathcal{C} \\ \mathcal{C} \text{ is Topology}}} \mathcal{C}$.

Thus $\mathcal{C}_\beta \subseteq \bigcap_{\substack{\beta \subseteq \mathcal{C} \\ \mathcal{C} \text{ is Topology}}} \mathcal{C}$, the inverse inclusion is clear since the Right term is: the smallest Topology containing β .

Lemma. Let X be a set, and $\mathcal{C}_1, \mathcal{C}_2$ be Topologies on X .

Then, $\mathcal{C} \stackrel{\text{def}}{=} \mathcal{C}_1 \cap \mathcal{C}_2$ is a Topology on X .

Proof. $\emptyset, X \in \mathcal{C}$ is clear.

Let $U_\alpha \in \mathcal{C}$ ($\alpha \in I$) be open sets. Then,

$$\bigcup_{\alpha \in I} U_\alpha \in \mathcal{C}_1, \mathcal{C}_2.$$

Thus closed under union. Similarly closed under intersection.

Def. Let $(X, \mathcal{A})$ be a top. sp., $A \subseteq X$ subset

$$\mathcal{A}_A \stackrel{\text{def}}{=} \{A \cap U \mid U \in \mathcal{A}\}$$

forms $(A, \mathcal{A}_A)$ as subspace.

Lemma. $(X, \mathcal{A})$ and subspace $(A, \mathcal{A}_A)$. For $V \subseteq A$,

$$1) V \in \mathcal{A}_A \Leftrightarrow \exists U \in \mathcal{A} \text{ s.t. } V = A \cap U.$$

$$2) V \text{ is closed in } A \Leftrightarrow \exists C \text{ closed set of } X \text{ s.t. } V = A \cap C.$$

Proof. 1) is clear.

$$\exists C \subseteq X \text{ s.t. } C \text{ is closed and } V = A \cap C.$$

$$\Leftrightarrow \exists U \in \mathcal{A} \text{ s.t. } V = A \cap (X \setminus U) = A \cap X \cap U^c = A \cap U^c$$

$$\Leftrightarrow \exists U \in \mathcal{A} \text{ s.t. } A \setminus V = A \cap (A \cap U^c)^c = A \cap (A^c \cup U) = A \cap U \in \mathcal{A}_A.$$

Thm. $(X, \mathcal{A})$, $B \subseteq A \subseteq X$.

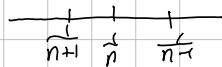
1) If $B \in \mathcal{A}_A$ and $A \in \mathcal{A}$, $B \in \mathcal{A}$.

2) If B closed in A and A closed in X , then B closed in X .

Counterexamples,

Let $(\mathbb{R}, \mathcal{A})$ be an Euclidean Space, and $K \stackrel{\text{def}}{=} \{\frac{1}{n} \mid n \in \mathbb{N}\}$.

Then, the subspace $(K, \mathcal{A}_K)$ is discrete Topology:



$$\text{for any } n \in \mathbb{N}, \left\{ \frac{1}{n} \right\} \cap \left(\frac{1}{n} - \frac{1}{n(n+1)}, \frac{1}{n} + \frac{1}{n(n-1)} \right) = \left\{ \frac{1}{n} \right\}.$$

Thus, every singleton in K is open in K , but not open in $\mathbb{R}$.

In fact, it is closed in $\mathbb{R}$.

Def. Let (X, τ) be a Topological space, and $A \subseteq X$ be a subset.

Define an interior of A in X as:

$$\text{int}_X(A) \stackrel{\text{def}}{=} \bigcup_{\substack{U \in \tau \\ U \subseteq A}} U$$

When the Ambient space is apparent, then denote $\text{int}_X(A) = A^\circ$.

Note: Trivially, A° is the largest open set contained in A .

Prop. Let (X, τ) be a Topological space, and $A, B, A_\alpha \subseteq X$ ($\alpha \in I$).

$$1) A \subseteq B \Rightarrow A^\circ \subseteq B^\circ$$

$$2) (A \cup B)^\circ \supseteq A^\circ \cup B^\circ$$

$$3) (A \cap B)^\circ = A^\circ \cap B^\circ$$

$$4) \left(\bigcap_{\alpha \in I} A_\alpha \right)^\circ \subseteq \bigcap_{\alpha \in I} A_\alpha^\circ$$

$$5) (A - B)^\circ \subseteq A^\circ - B^\circ$$

Proof. To show 1),

$$x \in A^\circ \Leftrightarrow \exists U \in \tau \text{ s.t. } x \in U \subseteq A \subseteq B \Rightarrow x \in B^\circ.$$

To show 2), using 1). Since $A, B \subseteq A \cup B$, $A^\circ, B^\circ \subseteq (A \cup B)^\circ$. So is the union.

Similarly to 2), $A \cap B \subseteq A, B$ implies $(A \cap B)^\circ \subseteq A^\circ \cap B^\circ$.

To show inverse inclusion, let $x \in A^\circ \cap B^\circ$. Then, there exist $U, V \in \tau$ s.t.

$$x \in U \subseteq A, \quad x \in V \subseteq B.$$

Now, $x \in U \cap V \subseteq A \cap B$. this means $x \in (A \cap B)^\circ$.

$$\text{Finally, for any } \gamma \in I, \quad \bigcap_{\alpha \in I} A_\alpha \subseteq A_\gamma \stackrel{1)}{\Rightarrow} \left(\bigcap_{\alpha \in I} A_\alpha \right)^\circ \subseteq A_\gamma^\circ.$$

$$\text{Thus } \left(\bigcap_{\alpha \in I} A_\alpha \right)^\circ \subseteq \bigcap_{\alpha \in I} A_\alpha^\circ$$

$$\text{5) is clear, being } (B^c)^\circ \subseteq (B^\circ)^c$$

Prop. Let (X, τ) be a Topological space, and $A \subseteq X$ be a subspace. For $B \subseteq A$,

$$\text{int}_X(B) = \text{int}_X(A) \cap \text{int}_A(B) \subseteq \text{int}_A(B).$$

Prop. If C is a collection of mutually disjoint open sets of $\mathbb{R}$, then C is countable.

Proof. Let $C = \{U_\alpha \mid \alpha \in A\}$ be a collection of mutually disjoint open sets.

For any $\alpha \in A$ and $x_\alpha \in U_\alpha$, there exists an open interval $[x_\alpha, x_\alpha] \subseteq U_\alpha$.

By density of Rationals, there is a $p_\alpha \in [x_\alpha, x_\alpha] \subseteq U_\alpha$.

Define a map:

$$f: C \rightarrow \mathbb{Q}: U_\alpha \mapsto p_\alpha.$$

Now, the disjointness gives f is injective.

That is, $f|_C: C \rightarrow f[C] \subseteq \mathbb{Q}$, C is countable.

Prop. Countable intersection of open sets of Co-countable space (X, τ_c) is open.

Proof. Let $U_i \in \tau_c$ ($i \in \mathbb{N}$) be open sets. Then,

$$\mathbb{R} \setminus \left(\bigcap_{n \in \mathbb{N}} U_n \right) = \mathbb{R} \cap \left(\bigcap_{n \in \mathbb{N}} U_n \right)^c = \mathbb{R} \cap \left(\bigcup_{n \in \mathbb{N}} U_n^c \right) = \bigcup_{n \in \mathbb{N}} \underbrace{\mathbb{R} \setminus U_n}_{\text{Countable}} \Rightarrow \text{Countable.}$$

Countable union

Def. Let $(X, \mathcal{U})$ be a Topological Space.

For a point $x \in X$, a set $N \subseteq X$ is said to be neighborhood of x if:

$$\exists U \in \mathcal{U} \text{ s.t. } x \in U \subseteq N$$

Denote $op(x)$ as set of open neighborhoods of x .

Def. Let $(X, \mathcal{U})$ be a Topological space, $A \subseteq X$ be a subset.

A point $x \in X$ is called limit point of A if:

$$\forall U \in op(x), A \cap (U \setminus \{x\}) \neq \emptyset.$$

Denote $A' \stackrel{\text{def}}{=} \{x \in X \mid x \text{ is limit point of } A\}$, and it is called derived set.

Prop. Let $(X, \mathcal{U})$ be a Topological space, A, B, A_α ($\alpha \in \Lambda$) be subsets.

$$1) A \subseteq B \Rightarrow A' \subseteq B'$$

$$2) (A \cup B)' = A' \cup B'$$

$$3) (A \cap B)' \subseteq A' \cap B'$$

$$4) \left(\bigcup_{\alpha \in \Lambda} A_\alpha \right)' \supseteq \bigcup_{\alpha \in \Lambda} A_\alpha'$$

Proof. At first, show 1). Let $x \in A'$. Then,

$$x \in A' \Leftrightarrow \forall U \in op(x), A \cap (U \setminus \{x\}) \neq \emptyset.$$

Now, since $A \subseteq B$, $A \cap (U \setminus \{x\}) \subseteq B \cap (U \setminus \{x\}) \neq \emptyset$. Thus $x \in B'$.

To show 3), $A \cap B \subseteq A, B \Rightarrow (A \cap B)' \subseteq A', B'$. clear.

To show 2),

$$\begin{aligned} x \in (A \cup B)' &\Leftrightarrow \forall U \in op(x), (A \cup B) \cap (U \setminus \{x\}) \neq \emptyset \\ &\Leftrightarrow \forall U \in op(x), (A \cap (U \setminus \{x\})) \cup (B \cap (U \setminus \{x\})) \neq \emptyset \\ &\Leftrightarrow x \in A' \cup B'. \end{aligned}$$

4) is clear by 1).

Def. Let (X, τ) be a Topological Space, and $A \subseteq X$ be a subset.

Define a Closure of A in X as:

$$\text{cl}_X A \stackrel{\text{def}}{=} \bigcap_{\substack{A \subseteq C \\ C \text{ closed}}} C.$$

$\lim_{x \rightarrow a}$

When the Ambient space is apparent, then denote $\text{cl}_X(A) = \bar{A}$.
Note: Trivially, $\bar{A}$ is the Smallest closed set containing A .

Lemma. Let (X, τ) be a Topological Space, and $A \subseteq X$ be a subset. TFAE:

a) $x \in \bar{A}$

b) For any open neighborhood U_x of x , $A \cap U_x \neq \emptyset$.

Proof. $\sim b) \Rightarrow \sim a)$:

$$\exists U \in \text{open} \text{ s.t. } A \cap U = \emptyset \Rightarrow \exists U \in \text{open} \text{ s.t. } A \subseteq \underbrace{U^c}_{\text{closed}}$$

$$\Rightarrow \exists U \in \text{open} \text{ s.t. } A \subseteq \bar{A} \subseteq U^c.$$

$$\Rightarrow x \notin \bar{A}.$$

Prop. Let X be a set, and $p \in X$. Define a Topology on X :

$$\mathcal{C} \stackrel{\text{def}}{=} \{U \subseteq X \mid p \in U\} \cup \{\emptyset\}$$

It is called *fore space* for p .

$$\{p\}^\circ = \{p\}, \quad \text{since } \{p\} \text{ is open.}$$

$$\{p\}' = \emptyset, \quad \text{since } \{p\} \setminus \{p\} = \emptyset.$$

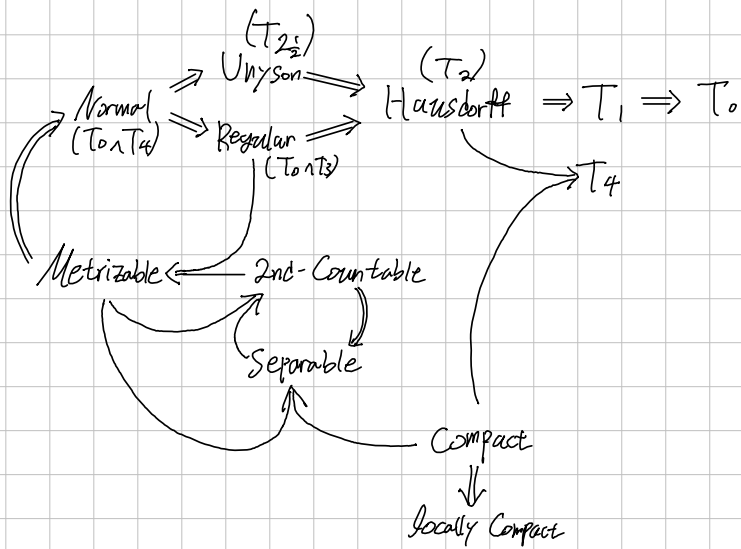
$$\overline{\{p\}} = \{p\}.$$

That is, $\{p\}$ is clopen.

$$\text{Let } q \in X \setminus \{p\}.$$

$$\{q\}^\circ = \emptyset,$$

$$\{q\}' =$$



[Metric Space.